Introduction

This reflective design diary overall documents the development I went through to create my **Bathroom Occupancy Detector** which is a smart home system created for shared bathrooms in student households however can be used widely in workplace environments too. The prototype I have created combines an Arduino UNO R4 WiFi, a PIR motion sensor and an LED indicator along with a web-based interface to answer the question of *“is the bathroom free?”* This project let me explore key concepts of IOT and helped me learn a lot about things like embedded logic, networking and human interface design on a small but very realistic scale.

The main goal for this Project that i had in mind was to build a working prototype that could detect motion and communicate this motion detection to an arduino that sends the data and information to a homemade website which will be made to display the information on any device both physically (with an LED indicator) and digitally (the local webpage). Overall this meant designing a hardware circuit with wires and a breadboard, implementing the arduino code that manages the PIR sensor input and the latched occupancy states, setting up wifi communication and creating a clear well displayed web interface. At the same time I needed to justify the design choices I had around usability, reliability and also privacy for the housemates that will live with the device.

This diary will first outline the aims, product vision and functional requirements that were needed to shape the project. It will then explain the design choices I had such as the sensor selection, occupancy logic and the communication methods while keeping the human centred issues in a shared bathroom in mind . The main sections in my diary will break down the design and development stages which include the circuit design , coding logic and diagrams with user interface and 3d design concepts. Finally at the end of the diary I will evaluate the outcomes and discuss the limitations of the project then identify future improvement given i had more time.

Aims, Product Vision and Requirements

Overall my aim for the project was to design and build a working bathroom Occupancy detector that solves a problem that me, my housemates and many housemates across the world have in shared housing, not knowing whether the bathroom is free until you physically go and check. The system I was going to make needed to be small and simple enough to fit in a small bathroom and also detect motion around the whole of that area where its placed. My aim was to cover as many IOT concepts as possible whilst doing this such as sensing, embedded logic, networking and interface. Rather than just building a half hearted useless Demo i wanted to create something that actually worked and solved these niche issues that people all over the world have and overall build something i was proud of. More specifically the aims for my project were.

- To detect bathroom Occupancy using a PIR motion sensor reliably
- To send the bathroom status from the Arduino UNO R4 WiFi to a web interface over wifi that i created to display necessary information and fun information.
- To display the status of the bathroom both physically with an LED indicator and digitally with the webpage that i created
- To justify the design choices in terms of how convenient it would be, the usability and privacy in bathrooms
- And finally to get information from my sensor, figure out a way to turn this information into valuable data through Arduino code, HTML code and javascript code then display this information on a webpage through Wifi

Product Vision

Here is a statement for my product vision:

“The bathroom Occupancy Detector uses a sensor and a webpage interface to show in real time if the shared bathroom is free or occupied so that people in a house or workplace don't have to waste their valuable time walking over or upstairs to check if the bathroom is free and then wait outside the door”

The key elements for the vision of my project would be to target a simple and small shared student house bathroom first and then extend to larger workplaces once completed.

User Stories

From my vision I defined some users' stories to put myself in other people's shoes so that I can get the most out of the project.

- As a housemate I want to see if the bathroom is free so I don't waste time walking upstairs and waiting outside for no reason.
- As a housemate i want the status to update automatically in real time so i dont have to refresh and keep checking the door
- As a housemate I want a simple but effective website that shows if the bathroom is free or occupied so I can check from anywhere in the house (additional information is a bonus)
- As a user i want an LED indicator outside the bathroom so i can see the status clearly without my phone
- As a student developer i want log occupancy changes so i have evidence that the system works and reflect on the usage patterns to display more cool information about the bathroom e.g busiest times, longest activity, shortest activity ect.

Functional Requirements

Here is where I converted the user stories into functional requirements for the project.

Must-Have

- PIR sensor that detects motion in bathroom
- Arduino UNO R4 WiFi converts sensor information to binary Occupied/Free state using code and connecting to PIR sensor using wires.
- Arduino acts as a server hosting the web page and displaying the simple `/status` function over WiFi
- Web page that gets the `/status` information and turns it into Bathroom Free or Bathroom Occupied
- An Led that mirrors the current state outside of the bathroom

Should-have

- Latched occupancy logic so that the system doesn't just flick between free and occupied or get stuck on occupied, instead hold the status for an average time that i calculated then checks again.
- Basic user interface for the web page (green for Free and Red for Occupied)
- Flexible web page so that the information can be displayed on any device size like a smaller iphone screen and still look presentable
- A simple connection indicator that tells me if the arduino and wifi/ webpage are connected so they can speak to each other and output the correct information

Nice-to-have

- A Notification when the bathroom is free/Occupied
- A Historical log of the recent occupancy changes
- A 3D printed model enclosure that fits the device perfectly
- Dynamic support for multiple bathrooms on one webpage dashboard

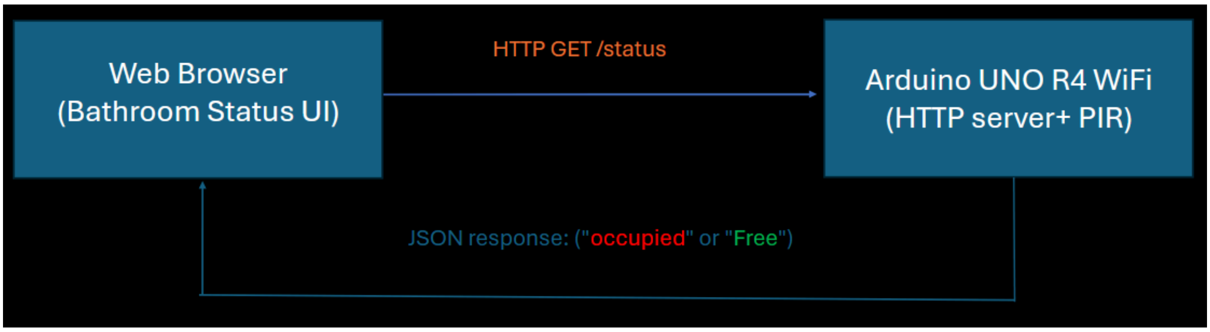
Design Justification

Sensor Choice (PIR sensor over more fancy options)

At the beginning of my project i considered combining multiple sensors (ultrasonic distance sensor with a magnetic door switch/Reed switch) to detect the occupancy more precisely. In the making of the project this added much more wiring complexity and more chance/opportunity to fail for not much overall gain. After testing out the HC-SR501 PIR i found out it reliably detects movement across the whole of my bathroom area and much more which i was initially very surprised about so i simplified my concept to the PIR sensor only and focused on reliable status logic instead. Overall for the project this reduced time, made the circuit smaller and was still very accurate enough for occupied/free logic along with more information that i could get through the sensor which i was very happy about.

Arduino/sensor communication method

There are many possible ways i could have gone around displaying the information from the arduino and sensor such as a cloud dashboard or mobile app however for a single bathroom on a home network this would be over the top. I used a local HTTP endpoint that returns a small JSON object with the current occupancy state. The website that i made uses `fetch(STATUS_URL, { cache: "no-store" }),` on a timer. This approach made it much easier to implement, easy to test back and forth with the browser and it also works on any device thats connected to the same WiFi. Below is a small diagram i made showing the Browser into `/status` to the Arduino UNO R4 then to JSON response



Small Ethical Considerations

Design choices were checked against how real people in a shared house would use the system. A single binary state (“free” or “Occupied”) avoids overall exposing peoples personal data when using the bathroom. However i did intend on implementing into the project many different usage analytics that the people in my house were okay with, this being Total visits, Last visit, Average Visit, Longest visit,Shortest visit and since last motion statistics. I wanted to add this as i had time to tweak the final outcome of my project and wanted to interesting and fun statistics to gain from the occupancy detector to really push this small simple PIR sensor and Arduino to see what the best i can get out of it is.

Design and Development Process

Process and Planning

For my process and planning I followed a simple Agile style process as a solo developer. I started by defining a clear product vision and a small set of user stories, then used these to build a backlog and a sprint goal to get a single bathroom working reliably from my sensor to the LED to the webpage. Each session I spent working on this project I noted what I planned, what I actually achieved and any issues that I came across for example WiFi connection problems,PIR sensitivity, and hardware malfunctions. Then I adjusted the next steps. This made sure as the developer of the project I didn't drift into endless ideas and focused purely on the task at hand to prioritise a stable end to end prototype.

My Personal Backlog

Must Have (Core Functionality)	Should Have (Important but not Essential)	Nice to have (Extras)
• A motion sensor that detects the occupancy of the bathroom • The Arduino sending the bathroom status over to the App/Webpage Via WIFI • Basic Web/App interface to show if the bathroom is Free/Occupied • An LED indicator for local Feedback incase all else fails and just for ease.	• A Door sensor to improve the accuracy (only occupied if the door is closed and motion is detected • Timer built into the code, a timer that ensures that the system doesn't display occupied when it is free (auto set "free" after 20 minuets no motion detected) • Basic Styling in the App or Webpage (green colours for free and red for Occupied).	• A notification when the bathroom is free • A funny message that may say "Occupied Someone is busy in there !" • Data logging (statistics) of the busiest times of that day or even the longest time of bathroom use.

Agile style backlog showing my product vision

Hardware Design (Circuit)

One of my first steps of the hardware design was to build and actually test the hardware:

Components: Arduino UNO r4 WiFi, HC-SR501 sensor, red LED, resistor , large breadboard and jumper wires.
Connections: PIR VCC → 5V, GND → GND, OUT → digital pin 2; LED anode → digital pin 3 via resistor, cathode → GND.

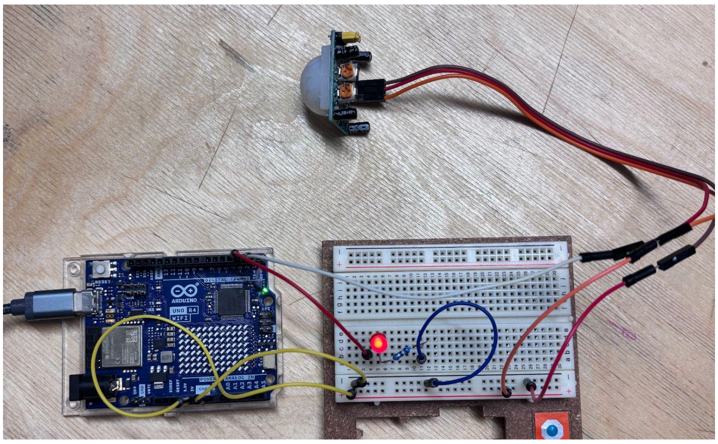


image of original hardware setup.

As seen above i initially tested on a very large breadboard to give space for experimenting with the different pin layouts and cable management and also to start with a good understanding of how the connections work as it was my first time using a breadboard and jumper wires so having a small compact breadboard would have been very tricky to figure out the first time round. After a long process of this project I confirmed that everything fit on a small breadboard. The reason I wanted a smaller breadboard was so that in the future I could add this system into a smaller enclosure with a more compact system later down the line. All of my early tests for this stage of the project were mainly walking in front of the PIR at different distances and angles to gauge the detection range/false triggers and overall make sure that the sensor would be enough to detect motion in the whole bathroom without any other necessary sensors having to be added to the project.

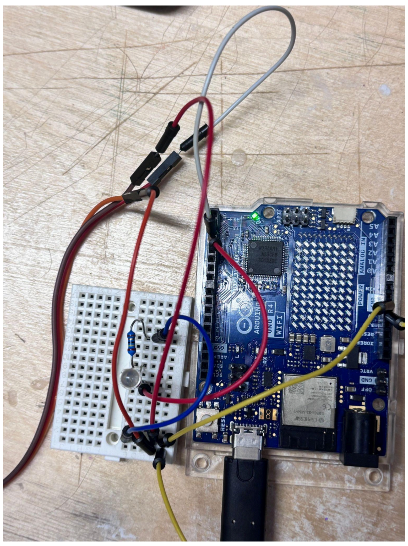


Image of updated smaller breadboard

Embedded Logic

Once i had the sensor and LED and everything else wired, i wrote the Arduino code to begin turning the raw motion into a stable occupied or free state. The PIR sensor gives short HIGH pulses when it detects motion so i introduced two key variables which are `bool isOccupied = false;` (boolean) and `unsigned long lastMotionMs = 0;` (unsigned long). On each loop this is what happens behind the code:

- If PIR is HIGH set `isOccupied = true;` and then update `lastMotionMs`

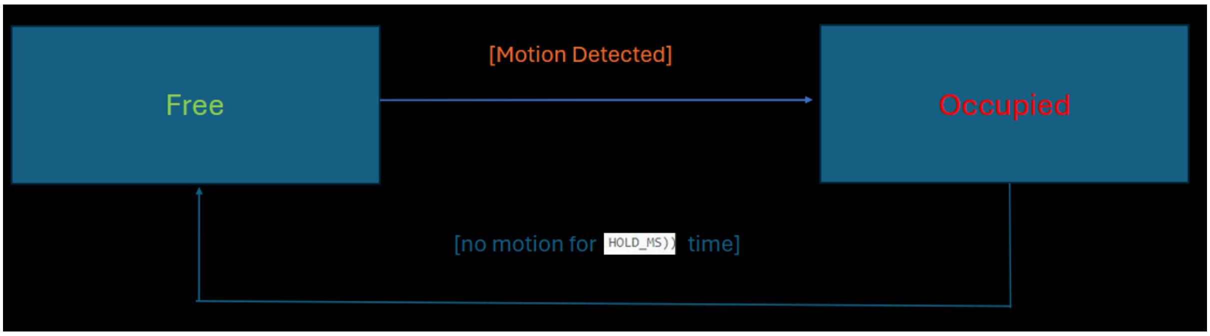
```
if (!isOccupied) {
  isOccupied = true;
  Serial.println("State -> OCCUPIED")
}
```

- If `isOccupied` is true and no motion has been seen for a set time `HOLD_MS))` is set to `isOccupied = false;`

```
if (isOccupied && (nowMs - lastMotionMs >= HOLD_MS)) {
  isOccupied = false;
```

- Then the LED state always mirrors the `isOccupied` boolean

This latched time approach that i created was made to stop a flickering issue i had with the system where the status used to just switch back and forth between occupied and free whenever someone moved in front of the sensor so it behaved more like a person in person out detector rather than a raw motion sensor. This was necessary for the project as if you checked on a device to see the bathroom status someone could be in there but it would say free and switch to occupied only when someone moves and switch straight back to free which was giving unreliable data.



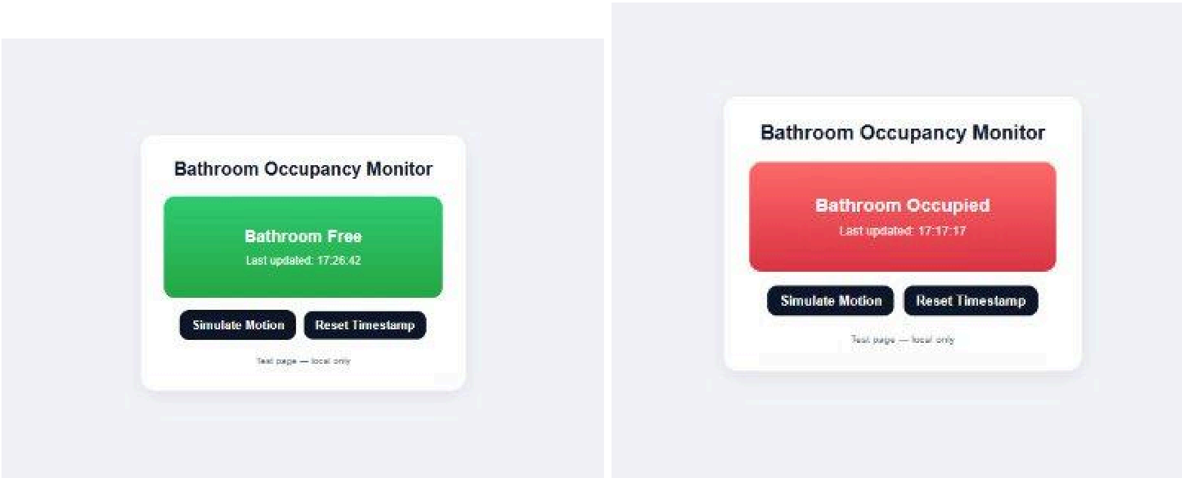
Web interface and WiFi integration

With all of the occupancy logic working and all of the hardware setup with arduino code i began to integrate WiFi using the WiFiS3 library and references from Github. The Arduino connects to the home network, starts a `WiFiServer server(80);` on port 80 and fetches `/status` the returning JSON such as

```
{ "occupied": true, "timestamp": "2025-11-24T14:32:10Z" }
```

Then on the front end of things a small java file runs in the browser and keeps on calling

```
fetch("http://ARDUINO_IP/status")
```



Old image of previous website design displaying Free/Occupied status

Analytics Features and Testing Data

As this project went on i wanted to add much more detail and interactivity to the whole project so to go beyond the simple on/off status (that you can see in the images above) i extended the code to track each bathroom visit and other basic analytics. Each time the system transitions from Free to Occupied and then Back to free it updates like this:

- TotalVisits - count of completed visits
- SinceLastMotion - the last time that the PIR motion sensor was triggered
- lastVisitDurationMs - The time spent on the most recent visit
- longestVisitMs - The longest time the Bathroom had been visited
- shortestVisitMs - The shortest time the bathroom had been visited
- AverageVisit- the average time that the bathroom gets visited

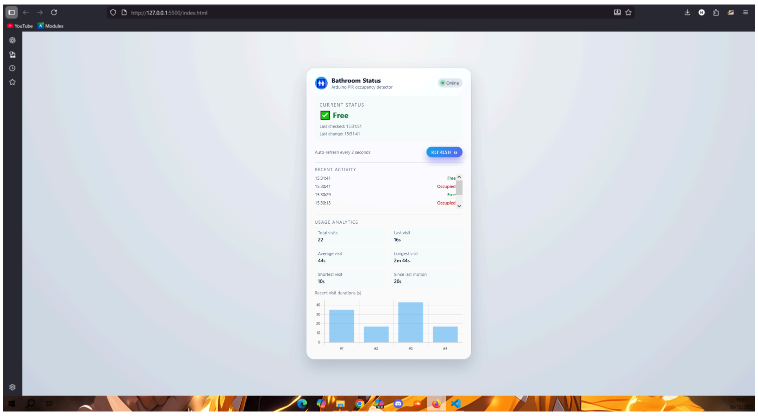
These values that i created are using millisecond based timestamps when a visit starts and ends, and are then added to the JSON response from `/status` using the information from the PIR sensor then debugging this information with my Arduino code to make the information useful and finally sending this into my js and HTML files to display the information.

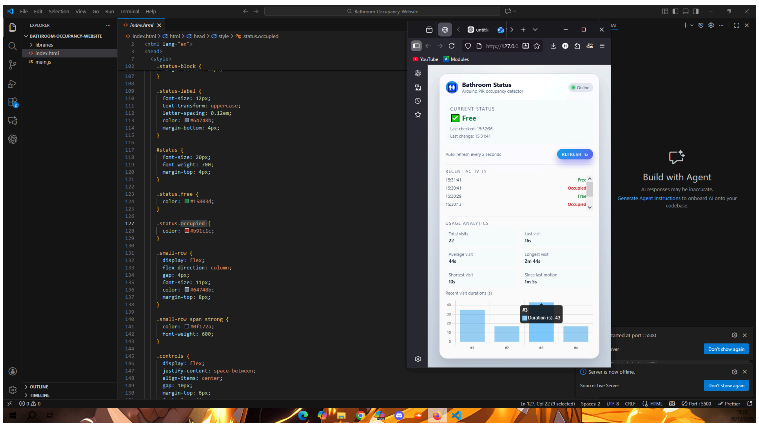
Statistics Variables

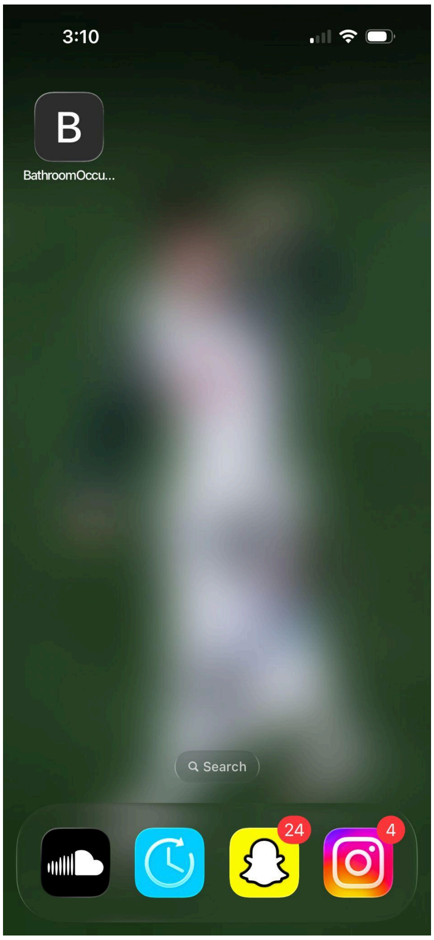
```
const statTotalVisits   = document.getElementById("stat-total-visits");
const statLastVisit     = document.getElementById("stat-last-visit");
const statAvgVisit      = document.getElementById("stat-avg-visit");
const statLongestVisit  = document.getElementById("stat-longest-visit");
const statShortestVisit = document.getElementById("stat-shortest-visit");
const statSinceLast     = document.getElementById("stat-since-last");
```

Mobile App and Standalone Implementation

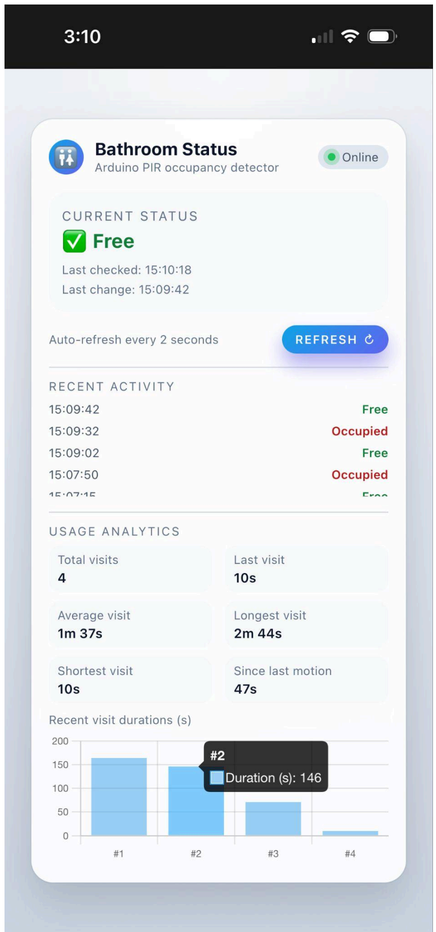
To get the most out of my website design I didn't want anyone who wants to use the application to have to log onto a laptop/PC device just to know whether or not the bathroom is occupied so I created an app that can be added to any device on the same WiFi as the Bathroom detector. You simply open the app and straight away in real time you can see the occupancy of your bathroom, all of the analytical data as shown, the recent activity of the bathroom and also a professional looking graph which lets you see all of the recent visits and how long they were. The app layout in which i created was optimised for any screen size and as you can see in the photos below it looks perfect on an iphone size and any laptop/desktop size and also works perfectly with resizing your screen .







Screenshots of working App inside iphone and evidence of Analytical data testing/resizing UI on all devices



Physical Enclosure (3D Design)

I began to start designing a model 3D printed enclosure in Tinkercad for the project to hold everything together. The internal volume would be the size of the arduino UNO R4 WiFi and the small breadboard (which i switched over to incase i had time for the 3D model) with a front cut out for the PIR sensor to stick out and also a top cut for my LED to stick out with a rear cut for the small usb-c cable. Also with this design i explored a simple hinged lid so the electronics can be accessed or maintained. Although i did not get enough time to print this enclosure, I worked hard creating it and thinking about the practical considerations like the sensor field of view, cable routing, mounting and water exposure. This is something in which i think would be the “icing on the cake” for my project and i would have liked to have completed this however i simply didnt have time to get this printed and i have plans of printing this even when the project is over and finished in my own time to better and improve my bathroom sensor for myself. As you can see in the image below i created the model and measured exactly the sizes for my breadboard, wires, Arduino and PIR sensor ready to print out just to show that if the time period was slightly longer this would have been done.

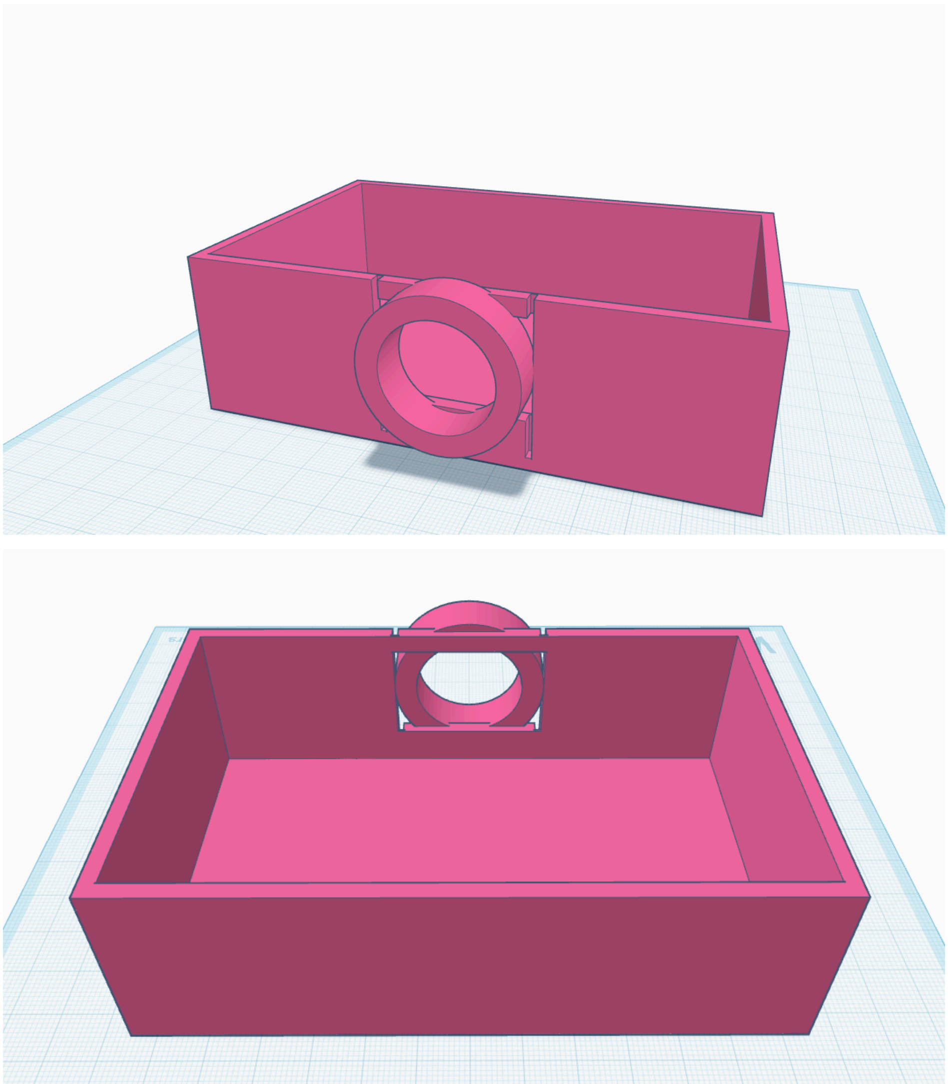


Image of 3D print out model enclosure prototype

Outcomes and Evaluation

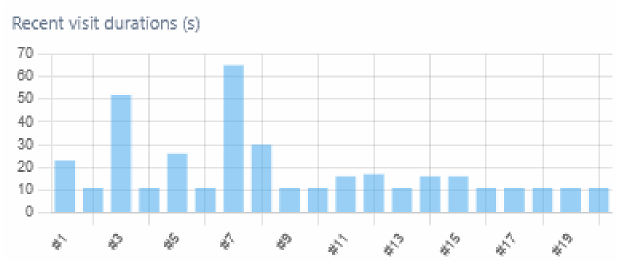
Functional Testing/ Analytics Testing

To test my project thoroughly I had to install this in the bathroom with a portable charger and leave the bathroom occupancy detector running throughout the day to track my housemates and my own movement in the bathroom. For each test iteration when somebody entered the bathroom I observed the LED , the web interface and the IPhone app simultaneously. The results that I came out with were much better than what I was expecting and came out perfectly. Throughout the day of testing with the occupancy detector setup I had 0 issues. The detector worked for over 5 hours sensing the detection of motion and holding this for 10 seconds while updating and checking for motion every 2 seconds providing my user interface with lots of data in the analytics section that were interesting to see. The system consistently switched to occupied in real time from the first motion event and remained in that state until somebody left the bathroom only returning to free after 10 seconds of the last motion detected with no further motion. Both the browser and app reflected the changes also with 0 errors confirming that the end to end pipeline behaves perfectly as it should do.

To fully test the analytics i noted start and end times manually for the bathroom visits that i had then compared these with the values that were reported by the system on my phone. The results that came through throughout the day were:

- Total visits: 14
- Average visit length: 2m 12s
- Longest visit: 6m 23s
- Shortest visit: 25 seconds

The systems values matched my notes within a few seconds of the range however if i tweaked the latched timings and delays slightly i could definitely get the timings within less than a second of the actual lengths. Any discrepancy that were to occur was expected because the PIR only triggers on movement and my code uses a timeout to infer when a visit ends however this was not really the case as the information came out with 0 issues. The key result in this information is that the analytics are based on real time Free/Occupied transitions rather than hard coded numbers, and they update live on all clients so i was very happy with the outcome of the testing. The only thing that i marked down as an error was that if someone entered the bathroom straight after someone else then the analytical data would have been wrong for that visit by around 8 seconds however this did not happen in my testing and i also think this is a minor error that can be easily ironed out.



*Recent visit Graph example using **Chart.js***

Usability and App Feedback

Testing with my housemates gave me the opportunity to ask for any feedback they had on my project. When asked both my housemates said that the interface was very easy to understand without any explanation. The combination of the LED light glowing through the door and large “ 🚫 Occupied / ✅ Free ” status on their phones meant that they could make sense of this instantly. The analytics were then seen as a fun extra to the project and also reassured both of my housemates and me that the detector was still updating.

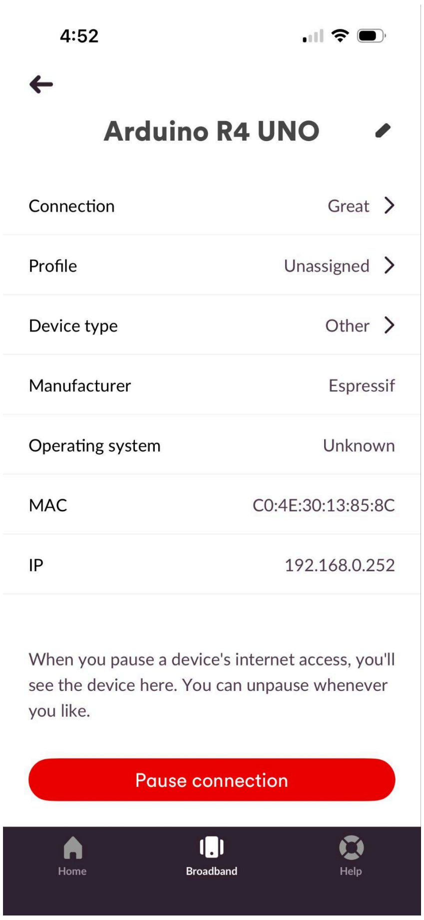


Image evidence of Arduino UNO R4 WiFi Connected to broadband

Image of LED indicator through the door

What Worked Well ✅

- Reliable end to end flow from PIR sensor through Arduino logic then to JSON and onto the web and mobile interfaces.
- Latched Occupancy logic that solved flicker and enabled useful analytics.
- The extra analytics that gave more insight from the same single sensor.
- Standalone use in the bathroom with portable charger and phone app, making my prototype feel like a realistic product.
- Overall Real time quick feedback of the Occupied/Free status
- Perfectly made working webpage and mobile app that resizes accordingly to the user

Limitations ❌

- The single PIR sensor with the timeout logic only approximates the occupancy so if the user stays very still for over 10 seconds this can cause minor miscounts (Very unlikely)
- Users who have Pets that enter the bathroom could cause miscounts
- Analytics are stored in the RAM so they reset if any power is lost to the Arduino meaning there is no multi day history.
- The system currently only supports one bathroom on a single local network and offers no secure remote access or multi bathroom dashboard
- The 3D enclosure is still just a model in tinkercad and has not been printed or tested physically yet

Reflection on Process

One of the big reflections I had through the process of building this was that simplifying the sensing to a single PIR sensor and focusing on a smaller working end to end product was crucial to my overall project, without that decision I would definitely have run out of time debugging multiple sensors. Once I had a stable working PIR it was easier to then focus on the harder parts like using the Arduino as its own server and communicating back and forth with not only the website HTML and JS code but also the local WiFi server.

I think that my overall planning for this project was good but not great. I did create a product vision, user stories and backlog however I drifted into spending crucial time doing the fun analytics tasks before fully completing and printing out the 3D model, even though I made the model and can just print this out I should've left time to test the model and make improvements to this. My project only really leveled up when I forced myself to treat all of the analytics and app integration as proper backlog items and not just last minute extras.

Throughout this process I learned lots about the Internet Of Things module and genuinely thoroughly enjoyed it, making me want to continue this in my own time despite aiming for marks and grades. One thing i learned with my product was that building a clean user interface definitely pays off as everything revolved around a `/status` JSON endpoint and a single occupied state so extending the system with the statistics and a phone app was a challenge I enjoyed. If I had hard coded logic into my user interface or mixed my networking and sensor logic together those late changes would have been a struggle. My takeaway for this would be that even doing a small student project separating different tasks (sensing,state change, API, UI) using problem decomposition makes the whole project more adaptable.

Future Work

If i had another sprint to continue this project, I would focus on four main things

Improve occupancy accuracy

- I would try to add a magnetic door sensor (REED switch) and then combine this with the PIR sensor so that the occupied only shows when the door is closed and then motion has been detected recently
- I would also experiment with different timeout values more thoroughly than i did and possibly use adaptive timeouts based on typical visit lengths to reduce false "Free" states if someone stays very still.

More persistent analytics

- I would log the visit data (the start time and end time durations) toi some sort of storage for example an SD card so that the statistics continue even when the Arduino is turned off and has no power. This will enable multi day analytics for much more fun statistics such as busiest bathroom times based on the data and also enables much more reliable data for averages.
- I would also like to build more complex visualisations with daily or weekly charts that would consist of number of visits and the total usage time/average time in the bathroom in order to turn the current fun facts I have into more serious usage stats.

Enclosure and Setup in a real bathroom

- I would have obviously liked to finalise and actually print/test the enclosure for the product and then mount the device inside an actual bathroom and run this continuously for several weeks to gain some better data along with the persistent analytics.
- Another thing I would like to test is how the product copes with real conditions like steam, temperature changes, people's behaviour and pets. This would be so that I can adjust sensor placement and timeout/LED visibility based on users feedback.

Scaling the bathroom system and enhancing the app

- It would be very efficient if i could extend the Arduino/API design so that multiple bathrooms can be monitored, each with its own ID and status. (e.g guest bathroom, upstairs bathroom, downstairs bathroom) or in a workplace (Office Bathroom, warehouse Bathroom, kitchen bathroom) all connected on one dashboard.
- I would also like to further update the mobile app to show a list of bathrooms and optionally send a notification to your device when the bathroom is occupied and when it also becomes free.
- The last thing I would like to improve as future work on this project would be to investigate secure remote access for example a VPN or a small backend so that users can check the status even when they are not connected to the same home or work wifi.

These changes would move the project from a single working prototype to a more robust and scalable IOT system that could definitely be used in a shared flat or small workplace.

References and Bibliography

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